MILITARY SPECIFICATION

SPRAYER, INSECTICIDE, ELECTRIC-MOTOR-DRIVEN, FRAME MOUNTED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a frame mounted, electric-motor-driven, insecticide sprayer.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

CC-M-636	Motor, Alternating-Current (Fractional Horsepower)

Military

MIL-S-12651	Sprayers, Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

(Copies of specification and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

FSC 3740

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicate, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN TRUCKING ASSOCIATION, INC. (ATA)

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P. Street, N. W., Washington, D. C. 20036.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 20	Standard for Snap Switches
UL 62	Standard for Flexible Cord and Fixture Wire

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611).

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-thread Standards for Federal Service

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Preproduction/First Article Test. This specification provides for preproduction/first article testing in Section 4.

3.2 Description. The insecticide sprayer shall consist of a power unit, dispensing system, gear pump, shut-off variable control nozzle, steel base, carrying handle, switch, and electrical cord.

3.3 Material. Material shall be as specified herein. Material not definitely specified shall be of the best quality used for the purpose in commercial practice. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

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3.4 Standard product. It is intended that the equipment and its component parts shall be regular commercial product of the manufacturer or his suppliers. The items furnished shall be new, unused, and free from defects and imperfection.

3.5 Interchangeability. For like equipment, all removable units and replacement parts shall be interchangeable with corresponding items or parts without requiring hand or machine fitting.

3.6 Design. Design of the equipment shall be in accordance with the best engineering practice. The equipment design and accessory installation shall provide accessibility for maintenance and service. The design shall be such as to prevent conditions which may be hazardous to personnel or deleterious to equipment.

3.7 Power unit. The power unit shall be an electric motor conforming to CC-M-636, type III, Class 7. Unless otherwise specified (see 6.2), the motor shall be rated at 115 volts and 1/2 horsepower. Fungus preventative treatment and radio interference suppression may be furnished, but is not required unless specified (see 6.2).

3.8 Dispensing system. The dispensing system shall consist of a shut-off variable control nozzle, an 8 ft neoprene supply line, an 8 ft neoprene return hose, a pressure gauge, a relief valve, and 50 ft of 3/8 inch neoprene spray hose. A removable strainer shall be provided in the supply line as required to prevent clogging of valve and spray mechanism. All parts shall be constructed, assembled, and supported to withstand testing in accordance with 4.5 without damage and impairment of function. Each internal part coming in contact with insecticide shall not change dimension, texture, or physical property, or show signs of chemical change, when tested in accordance with 4.5.5. The pressure relief valve and guage shall be located at the gear pump outlet and shall provide adequate relief for the gear pump when the nozzle is shut-off.

3.9 Gear pump. The pump shall be constructed of bronze and shall provide means for air tight connections to withstand testing in accordance with 4.5 without damage, or development of leaks. A means of lubrication shall be provided for all moving parts.

3.10 Shut-off variable control nozzle. The sprayer shall be equipped with an adjustable nozzle capable of being completely shut-off. The nozzle shall be capable of providing a spray and a solid stream when adjusted.

3.11 Steel base. The base shall be made of steel and be free of burrs and sharp edges. It shall keep the motor and pump drive in alignment at all times and shall be designed to withstand testing in accordance with 4.5.4 without damage or impairment of function.

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3.12 Carrying handle. The carrying handle shall be firmly attached in a convenient position for safe operation.

3.13 Switch. The switch shall be a toggle type and shall conform to Standard UL 20, with a rating not less than 8 amperes. The "OFF" and "ON" positions shall be clearly marked. The switch shall be conveniently located for safe operation.

3.14 Cord. The cord shall conform to Standard UL 62, type SO, three conductor, with a ground type plug cap, permissible maximum current capacity of 12 amps, and length of not less than 25 feet.

3.15 Performance. During steady operation, flow at the 100 psi setting shall not be less than 3.5 $\pm$ .25 gpm. Flow at medium or intermediate settings, as provided, shall be in proportion between the high and low settings.

3.16 Treatment and finish. Unless otherwise specified (see 6.2), the equipment shall be treated and finished in accordance with the manufacturer's best standard practice.

3.17 Technical publications. Repair parts list, and operation and maintenance instructions, as applicable, shall be furnished with each sprayer.

3.18 Identification marking. The equipment shall be marked for identification in accordance with MIL-STD-130.

3.19 Workmanship. Workmanship shall be in accordance with good commercial practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection lot. All sprayers offered for delivery at one time shall be considered a lot for purposes of inspection.

4.3 Sampling. A random sample of sprayers shall be selected from each lot in accordance with MIL-STD-105.

4.3.1 Sampling for examination. Examination shall be based on inspection. Examination shall be based on inspection level II and as Acceptable Quality Level (AQL) of 4.0 percent defective.

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4.3.2 Sampling for tests. Test shall be based on inspection level S-3 and AQL of 4.0 percent defective.

4.4 Examination. The preproduction sample, and each sample selected in accordance with 4.3.1 shall be examined for compliance with the requirements specified in section 3 of this specification. This element of inspection shall encompass all visual examinations and dimensional measurements.

4.5 Tests. The preproduction sample, when provided, shall be tested in accordance with 4.5.1 through 4.5.6. Each sprayer selected in accordance with 4.3.2 shall be tested in accordance with 4.5.1 through 4.5.3. When a preproduction sample is not provided, not less than one sprayer in each contract or order shall be tested in accordance with 4.5.1 through 4.5.6. Failure to pass any phase of any test shall be cause for rejection. Leaks, loss of insecticide or chemical solutions, evidence of electrolytic action and chemical changes, and any uncorrectable defect shall be cause for rejection of the inspection lot.

4.5.1 Discharge rate. During steady operation, with the pump outlet pressure set at 100 psi fill a container of a known volume. From the time taken to fill the container calculate the discharge rate. Failure to conform to 3.15 shall be cause for rejection. With flow at the high setting and nozzle shut-off, the pump and connections shall be checked for leaks. Any leaks shall be cause for rejection.

4.5.2 Pressure relief valve and gauge. During steady operation, with the nozzle shut-off and the relief valve closed as far as possible, there shall be no damage to the pressure gauge. Development of leaks or an incorrect pressure gauge reading shall be cause for rejection.

4.5.3 Shut-off nozzle. During steady operation, the nozzle shall be adjusted to obtain a spray and a steady stream of flow. Binding or jamming of the nozzle or development of leaks shall be cause for rejection.

4.5.4 Ruggedness. Three complete sprayers shall be dropped from a height of 3 feet onto a hardwood surface. The manner of dropping shall be such as to cause each sprayer to strike on entirely different positions in the three drops. The sprayer shall than be examined to determine damage, such as cracks which would affect operation of controls, cause leaks, loosen handle connections, break or misalign drive, spray mechanism, and motor bearing supports, expose electrically charged parts to accidental contact, and any other damage or distortion which could affect function in accordance with 3.15. The sprayers shall then be tested for discharge rate in accordance with 4.5.1. Ineffective performance shall be cause for rejection of the inspection lot.

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4.5.5 Insecticide resistance. After determination that the sprayer is in good operating condition, all replaceable and serviceable parts of the sprayer shall be disassembled. All parts which come in contact with insecticide, during normal operations, shall be immersed consecutively, for a period not less than 72 hours each, in each of the following test solutions:

(a) Three percent salt solution.

(b) Salt solution with 1 1/2 pounds per gallon dissolved dieldrin.

(c) Salt solution with 5 pounds per gallon malathion.

(d) Undiluted zylol.

(e) Undiluted deobase (number 9 oil).

The parts and the used test solutions shall be examined to determine evidence of electrolytic action and chemical action which might impair safe operation. The sprayer shall then be reassembled and tested in accordance with 4.5.1 through 4.5.3. Any damaged part prior to reassembly shall be cause for rejection and replacement of the part tested. Ineffective performance after reassembly shall be cause for rejection of the inspection lot.

4.6 Preparation for delivery inspection. The preservation, packaging, packing and marking of the sprayer shall be inspected to verify conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing. The preservation and packing shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The sprayers shall be preserved and packaged in accordance with level A of MIL-S-12651.

5.1.2 Level C. The sprayers shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. The packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The sprayers shall be packed in accordance with level A of MIL-S-12651.

5.2.2 Level B. The sprayers shall be packed in accordance with level B of MIL-S-12651.

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5.2.3 Level C. The complete sprayers shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Packing shall comply with UCC Uniform Freight Classification Rules or ATA National Motor Freight Classification Rules.

5.3 Marking. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The frame mounted electric-motor-driven sprayers is intended primarily for termite control.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, Number, and date of this specification.

(b) When preproduction sample is required (see 3.2).

(c) Voltage of electric power supply if other than specified (see 3.7).

(d) Fungus - preventative treatment and radio interference (see 3.7).

(e) When treatment and painting shall be other than specified (see 3.16).

(f) Repair parts and maintenance tools required (see 3.17).

(g) Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).

Custodians:
Army - ME
Navy - YD
Air Force - 84

Review Activities:
Army - ME
Navy - YD

Preparing Activity:
Air Force - 84

Project Number:
3740-0089

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE. AF-WP-O-MAY 64 505

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